Tapping / branch points

Buffers and clock gates often sit where the tree forks

The idea

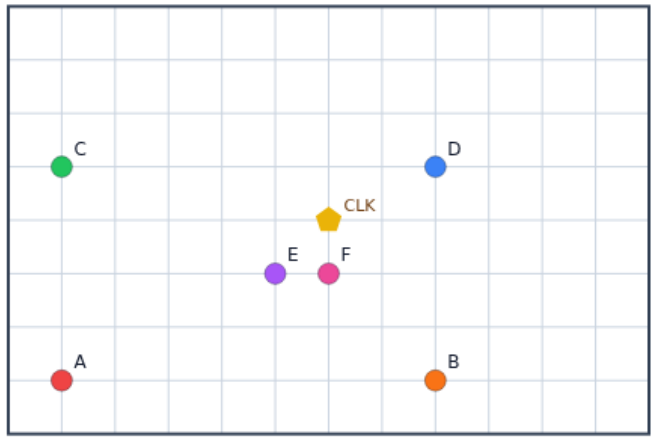
- Collect steiner and edge endpoints
- Keep points with degree at least three, those are tapping points

Branch taps

TAPPING POINTS

Branch taps

Step 1 / 5 · def · module02-07-tapping-points



Explanation

Tapping points are steiners with degree ≥ 3 — where the clock forks.

- Not leaf sinks
- Internal nodes

```
tappingPoints()
```

Branch taps

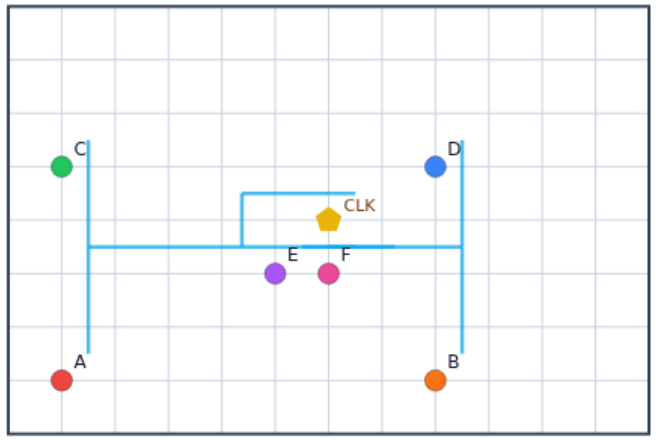
On MMM

TAPPING POINTS

On MMM

Step 2 / 5 · mmm · module02-07-tapping-points

skew: 1.50



Explanation

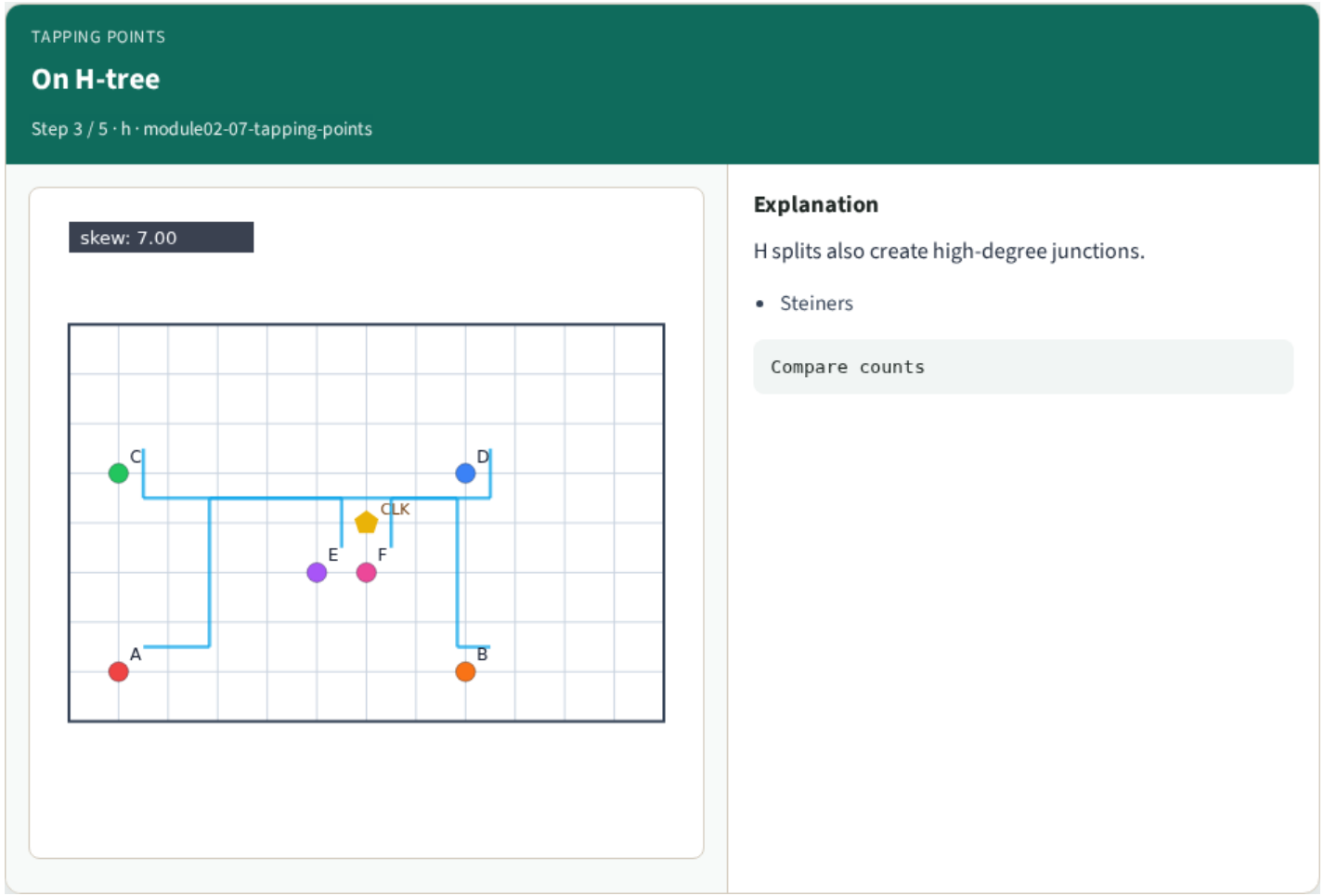
MMM merge points are natural taps.

- Count taps

```
taps≥1
```

On MMM

On H-tree



On H-tree

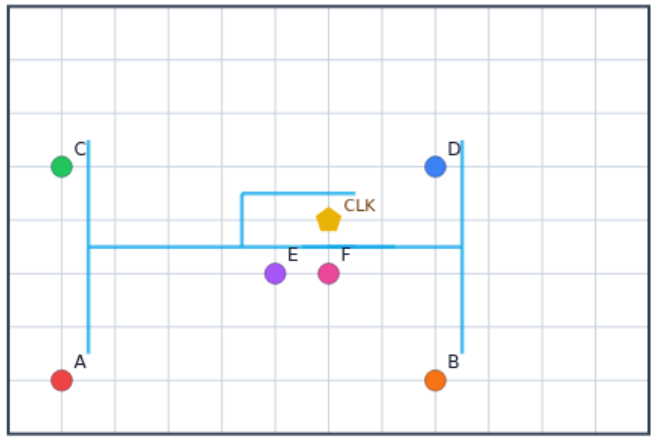
Why care

TAPPING POINTS

Why care

Step 4 / 5 · why · module02-07-tapping-points

skew: 1.50



Explanation

Buffers and gating often sit at taps; routing capacity concentrates there.

- Buffer sites
- Congestion

```
Clock buffering
```

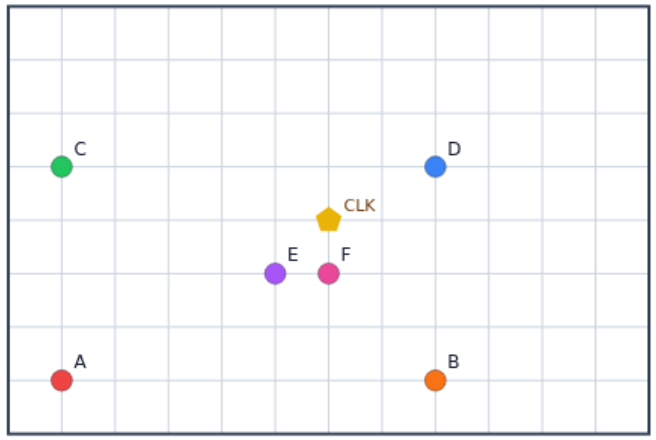
Why care

API

TAPPING POINTS

API

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Explanation

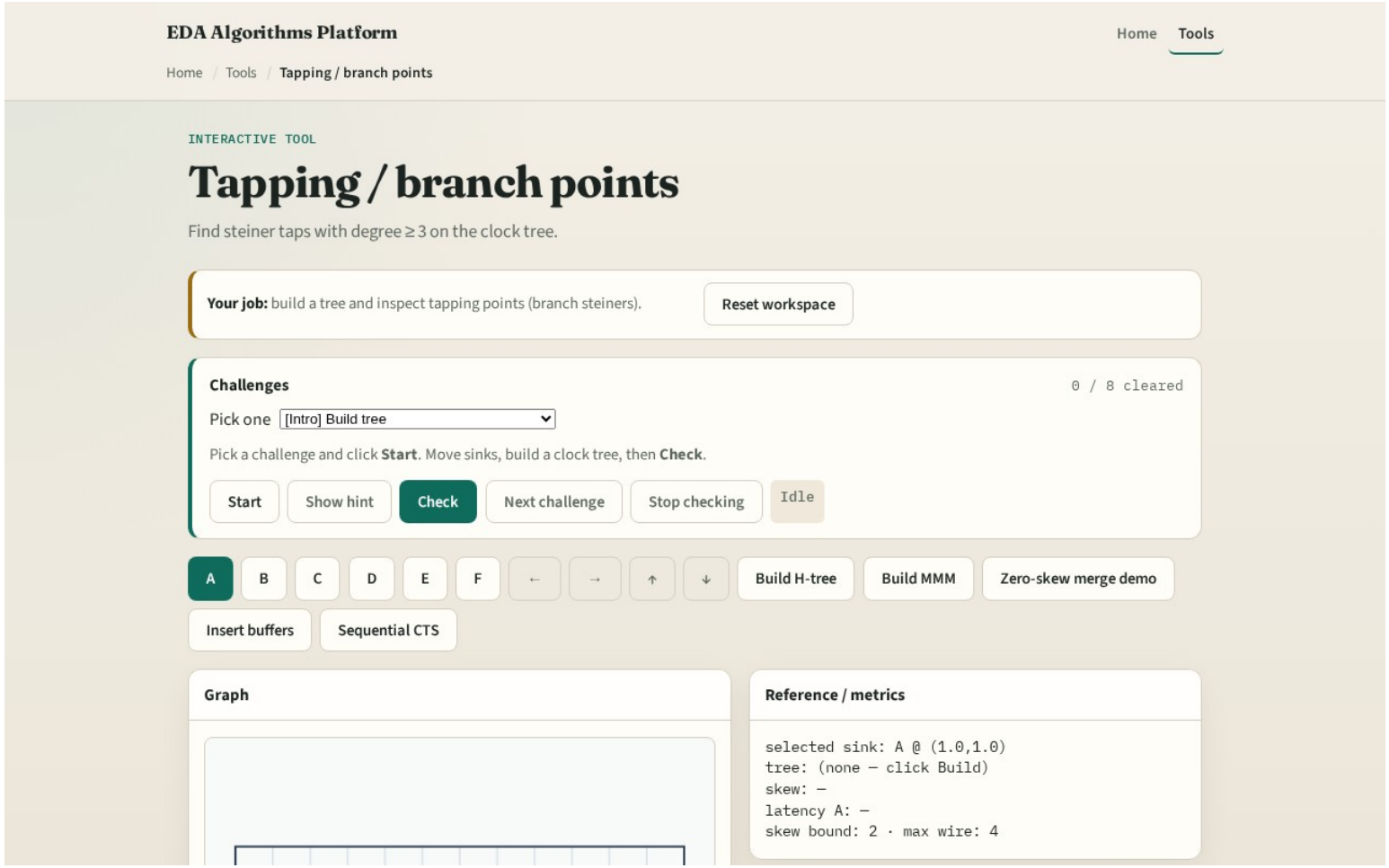
Call tappingPoints(tree) after Build — challenges score tap count.

- Learner tree
- Check

```
Next: buffers
```

API

Browser lab track



Browser lab starter

Implement track

- Call `tapping_points(tree)` after `mmm_tree`
- Assert length ≥ 1 on tiny_cts

Pitfalls

- Counting leaf sinks as taps
- Using degree on undirected edges inconsistently
- Dropping steiners before the degree pass

Your turn

- Next: insert buffers on long wires